

Statistical and Computational Models

Lecture 2: Linear regression as statistical learning

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Roadmap

Linear regression redux

RSS and the EPE

Two choices

Models tend to involve two choices – a choice of **function class** together with a choice of **loss**.

Function class: The type of functions that we are allowed to fit (e.g. linear functions, polynomials, step functions, trees).

Loss: What kind of 'error' matters, and how we price those errors.

The linear regression model combines:

- Functions of the form $\mathbf{x}^\top \boldsymbol{\beta}$
- Mean squared error loss

The design matrix

Returning to our voting example, linear regression requires an $n \times 1$ **response vector**, an $n \times (p + 1)$ **design matrix** whose first column is ones, and a $(p + 1) \times 1$ **coefficient vector**:

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}} \quad \underbrace{\begin{bmatrix} 1 & \text{age}_1 & \text{mail}_1 \\ 1 & \text{age}_2 & \text{mail}_2 \\ \vdots & \vdots & \vdots \\ 1 & \text{age}_n & \text{mail}_n \end{bmatrix}}_{\mathbf{X}} \quad \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}}_{\boldsymbol{\beta}}$$

Read: each row is an observation, each column is a predictor (and a column of ones for the intercept).

So $\mathbf{X}\boldsymbol{\beta}$ is $n \times 1$ – one fitted value per observation.

Question: what does the column of ones do? Allows the mean of $\mathbf{Y}$ to be non-zero when all covariates are zero.

Ordinary least squares

Choice one, the **class**: $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$.

Choice two, the **loss**: $L(\mathbf{y}, \hat{\mathbf{y}}) = \underbrace{(\mathbf{y} - \hat{\mathbf{y}})}_{\text{residual}}^2$

Now turn that loss into an **objective function** – sum it over the sample:

$$\text{RSS}(\mathbf{b}) = \sum_{i=1}^n (\mathbf{y}_i - \mathbf{x}_i^\top \mathbf{b})^2 = (\mathbf{y} - \mathbf{X}\mathbf{b})^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$$

And define our optimisation problem:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b}} \text{RSS}(\mathbf{b})$$

Read: find the $\mathbf{b}$ that minimises the objective, and so the summed loss.

Solving for $\hat{\beta}$

Step 1: expand the objective.

$$\text{RSS}(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b}$$

Step 2: differentiate with respect to $\mathbf{b}$, and set to zero.

$$-2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}, \quad \text{so} \quad \mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}$$

Read: the second equation is the **normal equations**.

Step 3: simplify, which requires $\mathbf{X}$ of **full column rank**.

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

QED

OLS as partialling out

Our regression has two features: age, and whether the modal voting type is by mail.

Suppose age is what we care about. Collect everything else – the constant, the mail indicator – into $\mathbf{Z}$.

1. Regress age on $\mathbf{Z}$. Keep the residuals, $\tilde{\mathbf{x}}$.
2. Regress $\mathbf{y}$ on $\mathbf{Z}$. Keep the residuals, $\tilde{\mathbf{y}}$.
3. Regress $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{x}}$.

Read: purge from both age and turnout everything $\mathbf{Z}$ can explain, then relate what is left over.

Partial interpretation

```
xt <- resid(lm(x ~ mail, data = train)) # age, mail projected out
yt <- resid(lm(y ~ mail, data = train)) # turnout, mail projected out

c(residualised = coef(lm(yt ~ xt))["xt"],
  full          = coef(lm(y ~ x + mail, data = train))["x"])
```

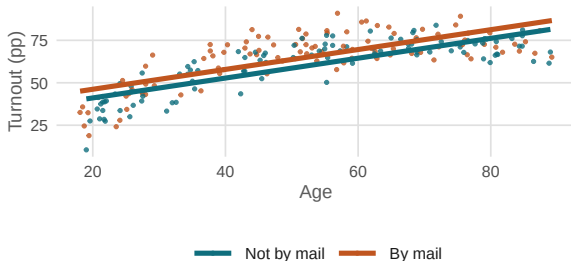
residualised	full
0.5848469	0.5848469

Read: every coefficient in a multiple regression is a **bivariate** regression of residualised Y on residualised X . This is the Frisch–Waugh–Lovell (FWL) theorem.

Residualising y is actually optional. $\tilde{\mathbf{x}}$ is already orthogonal to $\mathbf{Z}$, so whatever $\mathbf{Z}$ explains cannot reach $\hat{\beta}_{\text{age}}$.

Question: what if age is nearly a linear function of $\mathbf{Z}$? Then $\tilde{\mathbf{x}}$ is tiny, almost no variation in age survives, and $\hat{\beta}_{\text{age}}$ is mostly noise. This is ‘multicollinearity.’

Partial interpretation

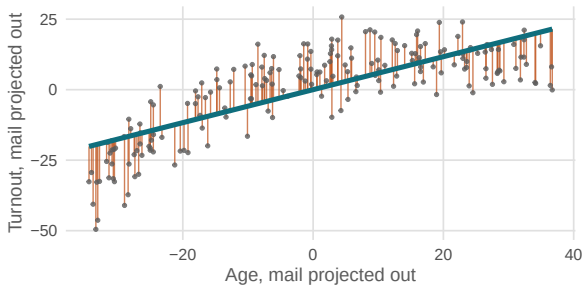


Raw observations, split by **Z**. One fit per group.

They are parallel by construction – that is what fitting **mail** as a shift means. Allowing different slopes on **age** would require an interaction term.

Question: what does partialling out **Z** remove? The gap between them.

Partial interpretation



Each point is one observation, with **Z** projected out of **both** axes (residualised).

The slope of that line is $\hat{\beta}_{\text{age}}$ – the number the multiple regression already gave us.

It is the line minimising the sum of the squared **residuals** (the vertical drops).

Roadmap

Linear regression redux

RSS and the EPE

Restricting the class

Recall that $f^* = \arg \min_f \text{EPE}(f)$; that is, f^* is the function, out of **every possible** one that minimises the EPE.

In linear regression we don't choose from every function, we choose from a **class of functions**:

$$f^\dagger = \arg \min_{f \in \mathcal{F}} \text{EPE}(f), \quad \mathcal{F}_{\text{lin}} = \{ \mathbf{x} \mapsto \mathbf{x}^\top \boldsymbol{\beta} \}$$

Read: $f^\dagger$ is the best function **in the class**, and $f^\dagger$ need not be f^* .

Note: The loss still fixes the **kind** of answer we get.

Three functions

f^* : the best function there is, but requires we know the DGP.

$f^\dagger$: the best function **in** $\mathcal{F}$, and still requires we know the DGP.

$\hat{f}$: the best function **in** $\mathcal{F}$ **given your sample**. This is what we estimate with OLS.

$$f^* \longrightarrow f^\dagger \longrightarrow \hat{f}$$

Read: OLS is two steps removed from f^* . The first comes from restricting the **class**, the second from using a **sample**.

The best linear predictor

$f^\dagger$ is linear, so a coefficient vector pins it down. Define $\beta^\dagger$ as the one minimising $\mathbb{E}[(Y - \mathbf{X}^\top \beta)^2]$. Solving that gives:

$$f^\dagger(\mathbf{x}) = \mathbf{x}^\top \beta^\dagger, \quad \beta^\dagger = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]^{-1} \mathbb{E}[\mathbf{X}Y]$$

This is the **best linear predictor**.

$$\hat{\beta} = \left(\frac{1}{n} \mathbf{X}^\top \mathbf{X} \right)^{-1} \left(\frac{1}{n} \mathbf{X}^\top \mathbf{y} \right)$$

Read: OLS is the **sample analogue**. Replace both expectations with sample averages.

Note: $\beta^\dagger$ exists whether or not $\mathbb{E}[Y | \mathbf{X}]$ is a straight line (here it is not). Many texts write it $\beta^\star$; we keep $\star$ for **unrestricted** optima.

Training vs. test vs. EPE

Lecture 1 ended with this table:

Quantity	Averages the loss over	Observable?
Training loss $\frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$	the n observations we fit $\hat{f}$ on	always
Test loss $\frac{1}{m} \sum_{j=1}^m L(y_j, \hat{f}(x_j))$	m observations not used in fitting $\hat{f}$	only if you have unseen data
EPE $\mathbb{E}[L(Y, \hat{f}(X))]$	the population (or very many draws from the DGP)	never

We want to find the function that minimizes EPE, but we **never** see the EPE.

Question: which row can we actually minimise? The first one.

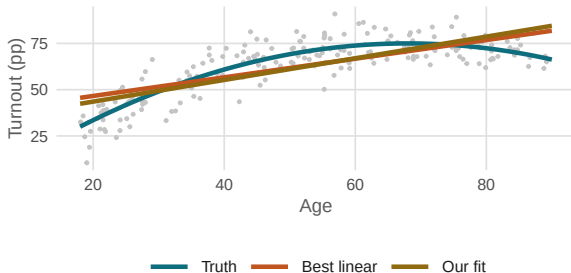
Substitute the one we can compute:

$$\text{MSE}_{\text{train}}(f) = \frac{1}{n} \sum_{i=1}^n \left(y_i - f(\mathbf{x}_i) \right)^2, \quad \hat{f} = \arg \min_{f \in \mathcal{F}} \text{MSE}_{\text{train}}(f)$$

Read: replace the expectation with a sample average, and minimise that instead.

$\text{MSE}_{\text{train}}$ is just RSS/n . So minimising RSS is minimising an **estimate** of EPE.

Three functions



Truth is f^* . **Best linear** is $f^\dagger$, fitted on the whole population. **Our fit** is $\hat{f}$, fitted on 200 observations.

Read: the gap between the first two comes from the class restriction. The gap between the last two comes from the sample.

Approximation and estimation

APPROXIMATION ERROR

$$\text{EPE}(f^\dagger) - \text{EPE}(f^\star)$$

From the **class** restriction.

Fixed once $\mathcal{F}$ is chosen, and will not shrink as $n \rightarrow \text{inf}$.

ESTIMATION ERROR

$$\text{EPE}(\hat{f}) - \text{EPE}(f^\dagger)$$

From **sampling**.

Not fixed, and shrinks as $n \rightarrow \text{inf}$.

In our case, the approximation error costs us **35.5** against a noise floor of **49**.

What consistency buys you

$$\hat{\beta} \xrightarrow{p} \beta^\dagger.$$

So $\hat{f} \rightarrow f^\dagger$, and the estimation error vanishes.

Question: does this get us to f^* ? No! Approximation error may remain.

References

- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **2.2, 3.1–3.2**. statlearning.com.
- Efron, B. & Hastie, T. (2016) *Computer Age Statistical Inference*, ch. 7.